

Submission STAT1-PS1-SUB02

Question STAT1-PS1-Q01

work / setup:

1. $\bar{x} = 33/5 = 6.60$

crossed-out arithmetic: [smudged]

FINAL (maybe): mean=6.60; sample variance=18.30 plus 1

Question STAT1-PS1-Q02

work / setup:

1. $P(D_1 \cap D_2) = (3/16)((3-1)/(16-1))$

crossed-out arithmetic: [smudged]

FINAL (maybe): 0.0250 plus 1

Question STAT1-PS1-Q03

work / setup:

1. $SE = \sqrt{(p(1-p))/n} = \sqrt{(0.35(1-0.35)/40)} = 0.0754$

2. $CI = 0.35 \pm 1.96(0.0754) = (0.202, 0.498)$

crossed-out arithmetic: [smudged]

FINAL (maybe): (0.202, 0.498); repeated-sampling interpretation plus 1

Question STAT1-PS1-Q04

work / setup:

1. Increase sample size n to reduce standard error.

2. Reduce confidence level or population variability when justified.

crossed-out arithmetic: [smudged]

FINAL (maybe): increase n ; lower confidence level or variability; note the coverage trade-off plus 1

Question STAT1-PS1-Q05

work / setup:

1. $0.03 < 0.05$, so reject H_0 under the stated rule.

crossed-out arithmetic: [smudged]

FINAL (maybe): reject H_0 ; evidence against H_0 , not proof that H_0 is false plus 1

Question STAT1-PS1-Q06

work / setup:

1. $\bar{x} = 21/5 = 4.20$

crossed-out arithmetic: [smudged]

FINAL (maybe): mean=4.20; sample variance=4.70 plus 1